Technical Reference

Temperature Controller API

May 7, 2025

BF1000-1234517327-58 Version 3.0, en-US
Original instructions

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Original instructions

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1 Introduction

1.1 Overview of the user instructions

This information applies to Bluefors Temperature Controller. There are different types of information products available for the product.

User Manual — Functional description of the product, component descriptions, and operating, maintenance, and troubleshooting instructions. Can also include installation and commissioning instructions.

Technical Reference — Necessary background information and technical details about a subject, such as parameter descriptions and use of scripts and API.

For details about the identification and location of the information products, refer to “Related information”.

These instructions are essential for the use of the Bluefors product. For safe and proper use of the product, read the instructions before use. Keep them for future reference.

1.2 Related information

Table 1: Related information

Information	ID	Location
Bluefors Temperature Controller User Manual	BF1000-1234517327-60	Available on Bluefors website: https://bluefors.com/support

1.3 Terms and abbreviations

Table 2: Terms and abbreviations

Term / Abbreviation	Definition
API	application programming interface
LAN	local area network
URL	unified resource locator

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1.4 Symbols and conventions

This section presents symbols and writing conventions that are used in the user instructions.

A note is used to indicate additional important information to the reader.

NOTE
Example of a note.

1.5 Customer service and support

Bluefors support includes reasonable telephone and email customer service during normal business hours (Finland). Support is provided by experienced technical personnel.

For support documents and downloadable software, refer to Bluefors Support website: https://bluefors.com/support.

For technical issues or questions related to the system operation, contact support@bluefors.com or +358 9 5617 4800.

For sales-related issues or questions, contact sales@bluefors.com or +358 9 5617 4800. For global sales contact information, refer to https://bluefors.com.

NOTE
In case of emergency or accidents, call your local emergency services.

1.6 Warranty

For warranty information, refer to the Bluefors warranty statement.

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2 API description

2.1 General information

This document describes the application programming interface for Bluefors Temperature Controller. The API is the interface for programs to access and control the device. Any programming language that supports REST API (HTTP GET/POST), WebSocket, or MQTT can be used.

2.2 Endpoint format

The API uses endpoints to access the data. The endpoints are URL patterns, such as network. For all API endpoint references, refer to “API reference”.

General rules related to endpoints:

- Format:
  - Element/sub-element/action
- The settings element is default and not written in the endpoints.
- The read action is default and not written in the endpoints.
- There are no IDs in endpoints.
- There are several main elements:
  - system
  - statemachine
  - channels and channel
  - heaters and heater
  - calibration-curves and calibration-curve

Examples of constructing endpoints:

- Reading /system settings
  - /system/settings/
- Subscribing to channels’ measurements
  - /channel/measurements/subscribe
- Read every calibration curve with full data
  - /calibration-curves/data/read

All interfaces are using two different modes for operation:

- request-response
- subscription.

For example:

- Request-response: */channel/measurement/latest* endpoint
  - Single request
  - Single response:
    - Response contains latest known data
- Subscription: /channel/measurement/listen endpoint
  - No request: Subscription done automatically by connecting to the endpoint (WebSocket) or subscribing to the topic (MQTT)
  - Unlimited number of responses:

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        - Every response contains the latest known data at that moment.

> **NOTE**
> By default, the WebSocket and MQTT interfaces are buffering unread responses to some limit defined by interface itself and next receive attempt could give the last unread response from that buffer.

2.3 Protocols

This section explains the supported protocols for the API.

2.3.1 MQTT

Connection:

- Port: 1883.

Topic formats:

- Incoming messages:
- endpoint/in
- Outgoing messages:
- endpoint/out

Client can operate in two modes:

- request-response
- subscription:
    - There is no specific subscription endpoint.
        - The same topic for outgoing messages is used for this

2.3.2 HTTP

Connection:

- Port: 5001.

URL formats:

- Supported:
    - http://host:port/endpoint
- Not supported:
    - http://host:port/endpoint/
    - http://host:port/endpoint/{id}
    - http://host:port/endpoint?id={id}
        - In this case all query parameters will be excluded from processing.

HTTP methods:

- GET

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- POST

Client can operate in one mode only:

- request-response

2.3.3 WebSocket

Connection:

- Port: 5002

URL formats:

- The same as for HTTP client.

Client can operate in two modes:

- request-response
- subscription:
    - Subscription done automatically by connecting to the endpoint.
    - Requests are not supported on this endpoint type.

2.4 Message payload

All message payloads are sent and received as JavaScript Object Notation (JSON) strings.

Format:

- JSON

Message payload also defines functionality in addition to endpoint:

- Read one/all elements
    - No payload
- Read one element
    - Payload with mandatory index fields
- Write one element
    - Payload with mandatory index fields and optional fields

2.4.1 Common fields

2.4.1.1 Request

Table 3: Request parameters

Key	Value type	Required	Comment
sender	string	n	Sender name. This information can be used by the request sender to recognize the response.

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Key	Value type	Required	Comment
hash	string (size: 0–50)	n	Original message hash number. This information can be used by the request sender to recognize the response.

NOTE
Request parameters are not possible in HTTP GET.

Example

```
{
  "sender": "client-name-or-id",
  "hash": "this-is-just-hash-number-issued-by-client",
  # varying data below, for example:
  "data1": ... # any data
  ...
  "dataN": ... # any data
}
```

2.4.1.2 Response

Table 4: Response parameters

Key	Value type	Required	Comment
sender	string	n	(the same as in request)
hash	string	n	(the same as in request)
datetime	datetime string	y	Datetime string in UTC format. Example: “2024-01-31T13:04:43.060313Z”
status	string	y	Allowed values: • “OK” • “ERROR”
error	error object	n	Always send if error

2.4.1.3 Error object

Table 5: Error object parameters

Key	Value type	Required	Comment
code	integer >=0	y	Error code
message	string (max 2000 chars)	y	Error message
details	string (max 2000 chars)	n	Additional information as a string

Example

```
{
  "sender": "client-name-or-id",
```


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```
  "hash": "this-is-just-hash-number-issued-by-client",
  "datetime": "2024-01-31T13:04:43.060313Z",
  "status": "ERROR",
  "error": {
    "code": 123,
    "message": "Wrong parameters provided.",
    "details": "Here are additional information as a string."
  },
  # varying data below, for example:
  "data1": ... # any data
  ...
  "dataN": ... # any data
}
```

2.4.1.4 List of error codes

Table 6: Error codes and descriptions

Code	Type	Name	Description
200	Error	No data	Internal error: not getting data from the measurement hardware.
241	Error	Signal too noisy	An indication of high resistance. Examine the sensor connection.
242	Error	Overflow	The measurement is saturated. An indication of the resistance being higher than the current excitation range.
243	Error	Common mode overflow	The measurements between the preamplifier ground (cryostat) and the temperature sensor lead are saturated. Indication of a noisy grounding.
244	Error	Zero measured value	Internal error: the device does not measure properly or there is a shorted circuit.
245	Error	Resistance too high	The resistance is above the maximum range.
246	Error	Resistance too low	The resistance is below the minimum range.
247	Error	Unreliable data	The measured signal arrives later than expected.
511	Warning	Temperature below range	The temperature is below the programmed impedance temperature curve.
512	Warning	Temperature above range	The temperature is above the programmed impedance temperature curve.

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Code	Type	Name	Description
514	Warning	Temperature below calibration range	The temperature is below the calibration range.
515	Warning	Temperature above calibration range	The temperature is above the calibration range.
518	Warning	Temperature data unreliable	The calculated temperature is negative.
519	Warning	No calibration curve	The calibration curve has not been defined.
521	Warning	Resistance below range	The resistance is below the calibration range.
522	Warning	Resistance above range	The resistance is above the calibration range.
901	Notification	Vmax auto-ranging	The current excitation is tuned to set the voltage to be the highest possible below the given value, Vmax.

2.5 Tutorial

2.5.1 Example: Access from browser

An easy way to check the device with an IP address 10.30.20.7110.0.0.135 which is connected to the Local Area Network is to use the browser. By typing the address http://10.0.0.13510.30.20.71:5001/system/network in the browser and pressing ENTER, this JSON response appears:


```
{"status": "OK", "state": "UP", "datetime": "2024-01-31T13:01:08.534671Z",
"netmask": "255.255.0.0", "dns": "", "mac_address": "b8:27:eb:fd:a4:00",
"ip_configuration": "DHCP", "ip_address": "10.0.0.135", "gateway": ""}
```


In the same way, when entering the address http://10.30.20.7110.0.0.135:5001/channel/measurement/latest, the latest measurement results appear:


```
{"rez": 10002.996333333334, "status": "OK", "angle": 89.59977777777777,
"temperature": null, "timestamp": 1572353805.343614, "resistance":
10003.483555555556, "channel_nr": 6, "datetime": "2024-01-
31T12:57:19.791712Z", "reactance": 1433539.4997777776, "magnitude":
10003.24, "status_flags": [511], "imz": 69.80922222222222}
```


NOTE

Responses are different depending on the status of the device.

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2.5.2 Example: Using an HTTP interface

In this example, the latest measurement results are requested programmatically from the IP address 10.30.20.7110.0.0.135 using Python language and the requests library. For more information about the library, refer to https://pypi.org/project/requests/.

```
# Import needed libraries
import json
import requests

# Actual request
req = requests.get('http://10.0.0.135:5001/channel/measurement/latest',
timeout=10)
data = req.json()
print('Response: \n{}'.format(json.dumps(data, indent=2)))
```

The printed result is:

```
Response:
{
  "rez": 10003.664111111111,
  "status": "OK",
  "angle": 89.6018888888889,
  "temperature": null,
  "timestamp": 1572355882.357602,
  "resistance": 10004.145555555557,
  "channel_nr": 6,
  "datetime": "2024-01-31T13:31:30.926461Z",
  "reactance": 1442043.1812222223,
  "magnitude": 10003.904888888888,
  "status_flags": [
    511
  ],
  "imz": 69.40755555555556
}
```

2.5.3 Example: Using a WebSocket interface

In this example, the latest measurement result is listened programmatically from the IP address 10.30.20.23110.0.0.135 using Python language and the websocket-client library. For more information about the library, refer to https://pypi.org/project/websocket_client/.

```
import json
import websocket

# Actual request
ws =
websocket.create_connection('ws://10.0.0.135:5002/channel/measurement/list
en', timeout=10)
resp = ws.recv()
data = json.loads(resp)

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))
```


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2.5.4 Example: Using an MQTT interface

In this example, the latest measurement results are subscribed programmatically from the IP address 10.30.20.23110.0.0.135 using Python language and the paho-mqtt library. For more information of about the library, refer to https://pypi.org/project/paho-mqtt/.

```
import json
import paho.mqtt.subscribe as subscribe
from time import sleep


# Callback to be called when new message received
def callback_print(client, userdata, msg):
    data = json.loads(msg.payload)
    print('Message: \n{}'.format(json.dumps(data, indent=2)))

subscribe.callback(callback_print, 'channel/measurement/listen',
hostname='10.0.0.135')

# infine loop of the actual program
while True:
    sleep(1)
```


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3 API reference

3.1 System

Table 7 shows the endpoint reference used to get system information.

Table 7: Endpoint: System information

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/system	/system	-	/system	system/in system/out

3.1.1 Read request

Empty.

3.1.2 Response

Table 8: Response information

Key	Value type	Required	Comment
api_version	integer	y	API version Allowed values: • 1
type	string (size: 0–50)	y	System type Allowed values: • “Bluefors TC”
serial	string (size: 0–50)	y	Device serial number
label	string (size: 0–30)	y	System label

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Key	Value type	Required	Comment
addinfo	string (size: 0–50)	y	System additional information
software_version	string (size: 0–50)	y	Software version

3.2 System / device

Table 9 shows the endpoint reference used to get device information.

Table 9: Endpoint: Device information

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/system/device	/system/device	-	/system/device	system/device/in system/device/out

3.2.1 Read request

Empty.

3.2.2 Response

Table 10: Response information

Key	Value type	Required	Comment
device_id	string (size: 0–32)	y	Device ID
device_firmware	string (size: 0–50)	y	Device firmware version

3.3 System / network

Table 11 shows the endpoint references used to read and write network settings. Only IPv4 is supported.

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Table 11: Endpoint: Network settings

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/system/network	/system/network		/system/network	system/network/in system/network/out
write	/system/network/update	-	/system/network/update	/system/network/update	system/network/update/in
subscription	/system/network/listen	-	-	/system/network/listen	system/network/listen

3.3.1 Read request

Empty.

3.3.2 Write request

Table 12: Write request information

Key	Value type	Required	Comment
ip_configuration	string	y	Allowed values: • "static" • "DHCP"
ip_address	string	n	IPv4 address
netmask	string	n	IPv4 address
gateway	string	n	IPv4 address
dns	string	n	IPv4 address

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3.3.3 Response

Table 13: Response information

Key	Value type	Required	Comment
<same fields as in write request>			
state	string	y	Possible values: • “UP” • “DOWN”
mac_address	string	y	MAC address

3.4 System / reset

Table 14 shows the endpoint reference used to enable system reboot or shutdown notifications. Only subscription is supported.

Table 14: Endpoint: System reboot / shutdown notification

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
subscription	/system/reset/listen	-	-	/system/reset/listen	system/reset/listen

3.4.1 Response

Table 15: Response information

Key	Value type	Required	Comment
message	string	y	Allowed values are: • “restart” • “shutdown”

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3.5 System / resources

Table 16 shows the system reboot / shutdown notification for the endpoint reference used to get information regarding OS resources. Only subscription is supported.

Table 16: Endpoint: System reboot / shutdown notification

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
subscription	/system/resources/listen	-	-	/system/resources/listen	system/resources/listen

3.5.1 Response

Table 17: Response information

Key	Value type	Required	Comment
uptime	integer	y	Uptime in seconds
memory_free	integer	y	Memory: free in Kb
memory_used	integer	y	Memory: used in Kb
cpu_total	float	y	CPU usage: total (in percentage)
disk_usage_log	integer	y	Disk usage: log disk (in percentage)
disk_usage_data	integer	y	Disk usage: data disk (in percentage)
rtc_battery	float	y	RTC battery: voltage

3.6 Statemachine

Table 18 shows the endpoint references used to read and write the settings of the measurement statemachine.

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Table 18: Endpoint: Measurement state machine settings – read / write

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/statemachine	/statemachine	-	/statemachine	statemachine/in statemachine/out
write	/statemachine/update	-	/statemachine/ update	/statemachine/update	statemachine/update/in statemachine/update/out
subscription	/statemachine/listen	-	-	/statemachine/listen	statemachine/listen

3.6.1 Read request

Empty.

3.6.2 Write request

Table 19: Write request information

Key	Value type	Required	Comment
wait_time	float: 1...100	n	Default wait time after changing channel before using values for measurement result. Value in seconds.
meas_time	float: 1...100	n	Measurement time for measurement operation. Value in seconds.
control_algorithm	integer: 0...100	n	Next channel selection algorithm: • 1: round robin

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3.6.3 Response

Table 20: Response information

Key	Value type	Required	Comment
<same fields as in write request>			
state	string	y	Textual representation of statemachine status
measuring	boolean	y	Measurements status: ON or OFF
channel_nr	integer: 1.. 12	y	Current measurement channel or last used channel

3.7 Channels

Table 21 shows the endpoint reference used to read all channel settings.

Table 21: Endpoint: Channel settings – read only

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/channels	/channels	-	/channels	channels/in channels/out

3.7.1 Read request

Empty.

3.7.2 Response

There is an array of results for each channel. Each element uses the same response format.

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3.8 Channel

Table 22 shows the endpoint references used to read and write channel settings.

Table 22: Endpoint: Channel settings – read / write

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/channel	-	/channel	/channel	channel/in channel/out
write	/channel/update	-	/channel/update	/channel/update	channel/update/in channel/update/out
subscription	/channel/listen	-	-	/channel/listen	channel/listen

3.8.1 Read request

Table 23: Read request information

Key	Value type	Required	Comment
channel_nr	integer: 1...12	y	Channel number

3.8.2 Write request

Table 24: Write request information

Key	Value type	Required	Comment
channel_nr	integer	y	Channel number (the same as in read request)
active	boolean	n	Active status: ON or OFF
name	string	n	Channel name

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Key	Value type	Required	Comment
excitation_mode	integer	n	Excitation modes: • 0: current excitation • 1: VMAX • 2: CMN
excitation_current_range	integer	n	Settings number for current excitation mode: • 1...22
excitation_cmn_range	integer	n	How much current is used for CMN excitation • 1: 50 uA • 2: 150 uA
excitation_vmax_range	integer	n	How much is the maximum voltage over the sample in VMAX mode • 1: 20 µV • 2: 200 µV
use_non_default_timeconstants	boolean	n	Possible values: • True: Use the channel's own specific wait and measurement times defined below. • False: Use default timers' values
wait_time	float: 1...100	n	How long time is waited after switching channel before doing the actual measurement in the channel. Value in seconds. Valid if use_non_default_timeconstants == True.
meas_time	float: 1...100	n	How much time is used to do actual measurement of single value in the channel. Value in seconds. Valid if use_non_default_timeconstants == True.
calib_curve_nr	integer: 1...100	n	The calibration curve used by this channel

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3.8.3 Response

Table 25: Response information

Key	Value type	Required	Comment
<same fields as in write request>			
coupled_heater_nr	integer: 0...4	y	Which heater is coupled to the channel: • 1...4: heater 1...4 • 0: no heater To change the relation, refer to “Channel / heater”.

3.9 Channel / historical data

Table 26 shows the endpoint reference used for getting a channel’s historical data from the device.

Table 26: Endpoint: Channel historical data

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/channel/historical-data	-	/channel/historical-data	/channel/historical-data	channel/historical-data/in channel/historical-data/out
read	/channel/historical-data	-	/channel/historical-data	/channel/historical-data	channel/historical-data/in channel/historical-data/out
read	/channel/historical-data	-	/channel/historical-data	/channel/historical-data	channel/historical-data/in channel/historical-data/out

3.9.1 Read request

Table 27: Read request information

Key	Value type	Required	Comment
channel_nr	integer: 1...12	y	Channel number

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Key	Value type	Required	Comment
start_time	datetime string	y	Possible data formats are: • YYYY-MM-DD • YYYY-MM-DD HH:MM • YYYY-MM-DD HH:MM:SS • YYYY-MM-DDTHH:MM:SSZ
stop_time	datetime string	y	Same data formats as for start_time
fields	array of strings	y	Requested data fields (channel): • “temperature” • “resistance” • “reactance” • “rez” • “Imz” • “magnitude” • “angle” • “timestamp” • “status_flags” • “channel_nr” For more information, refer to “Response”.

3.9.2 Response

Table 28: Response information

Key	Value type	Required	Comment
<same fields as in read request>			
measurements	object with data arrays	y	Requested data Names of data array elements are the same as field names.

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Key	Value type	Required	Comment
over_limit	boolean	y	Flag showing that amount of requested data is too big. If the over limit 100 000 is exceeded, no data is given.

Example

```
{
  "sender": "client-name-or-id",
  "hash": "this-is-just-hash-number-issued-by-client",
  "datetime": "2024-01-31T13:04:43.060313Z",
  "status": "OK",
  #
  "channel_nr": 1,
  "over_limit": False,
  "fields": [
    "timestamp",
    "temperature"
  ],
  "measurements": {
    "timestamp": [
      1542118174.989,
      ...
    ],
    "temperature": [
      0.034756,
      ...
    ]
  }
}
```

3.10 Channel / measurement

Table 29 shows the endpoint references used to get the latest real-time measurement data.

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Table 29: Endpoint: Latest real-time measurement data

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/channel/measurement/latest	/channel/measurement/latest	/channel/measurement/latest	-	-
subscription	/channel/measurement/listen	-	-	/channel/measurement/listen	channel/measurement/listen

NOTE

If older measurement data is needed, use the “historical-data” endpoint described in “Channel / historical-data”.

3.10.1 Response

Table 30: Response information

Key	Value type	Required	Comment
channel_nr	integer: 1...12	y	Channel number
resistance	float | empty	y	Resistance (Ohm)
reactance	float | empty	y	Reactance (Ohm)
temperature	float | empty	y	Temperature (K)
rez	float | empty	y	Real part of impedance (Ohm)
imz	float | empty	y	Imaginary part of impedance (Ohm)
magnitude	float | empty	y	Magnitude of impedance (Ohm)
angle	float | empty	y	Angle of impedance (degrees)
timestamp	float	y	End time of measurement (in Unix time)
settings_nr	integer: 1...24	y	Measurement settings number
status_flags	array of integers	y	List of measurement error codes. Refer to “Table 6”.

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3.11 Channel / heater

Table 31 shows the endpoint reference used to change the relation between a channel and a heater.

Table 31: Endpoint: Relation between a channel and a heater

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
write	/channel/heater/update	-	/channel/heater/update	/channel/heater/update	channel/heater/update/in channel/heater/update/out
subscription	/channel/heater/listen	-	-	/channel/heater/listen	channel/heater/listen

3.11.1 Write request

Table 32: Write request information

Key	Value type	Required	Comment
channel_nr	integer: 1...12	y	Channel number
heater_nr	integer: 0...4	y	Heater number

3.11.2 Response

Table 33: Response information

Key	Value type	Required	Comment
relations	array of ChannelHeaterRelation objects Refer to “ChannelHeaterRelation object”.	y	Refer to the definition below.

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3.11.3 ChannelHeaterRelation object

Table 34: ChannelHeaterRelation object information

Key	Value type	Required	Comment
channel_nr	integer	y	Channel number
heater_nr	integer	y	Heater number

3.12 Heaters

Table 35 shows the endpoint reference used to read all heaters settings.

Table 35: Endpoint: Heater settings – read

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/heaters	/heaters	-	/heaters	heaters/in heaters/out

3.12.1 Read request

Empty.

3.12.2 Response

There is an array of results for each heater. Each element uses the same response format as shown in Table 40.

3.13 Heater

Table 36 shows the endpoint reference used to read and write heater settings.

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Table 36: Endpoint: Heater settings – read / write

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/heater	-	/heater	/heater	heater/in heater/out
write	/heater/update	-	/heater/update	/heater/update	heater/update/in heater/update/out
subscription	/heater/listen	-	-	/heater/listen	heater/listen

3.13.1 Read request

Table 37: Read request information

Key	Value type	Required	Comment
heater_nr	integer: 1...4	y	Heater number

3.13.2 Write request

Table 38: Write request information

Key	Value type	Required	Comment
heater_nr	integer	y	Heater number (the same as in read request)
active	boolean	n	Active state: ON or OFF
name	string	n	Heater name
pid_mode	integer	n	Heater mode: • 0: Manual mode • 1: PID mode
resistance	float: 0...1000	n	Heater resistance in Ohms
power	float: 0.0...1.0	n	Applied manual power in Watts. The maximum power is 100 mA.

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Key	Value type	Required	Comment
max_power	float: 0.0...1.0	n	Hard safety limit for power in Watts
target_temperature	float: 0...1000.0	n	Manual mode: target temperature
target_temperature_shown	boolean	n	Manual mode: show target temperature in the graph
control_algorithm	integer	n	Control algorithms: • 1: Conventional (default) • 2: Step
control_algorithm_settings	ControlAlgorithmSettings object	n	Control algorithm settings
setpoint	float: 0...1000.0	n	Set point for control algorithm

3.13.3 ControlAlgorithm settings

Table 39: ControlAlgorithm settings

Key	Value type	Required	Comment
proportional	float	n	PID P
integral	float	n	PID I
derivative	float	n	PID D

3.13.4 Response

Table 40: Response settings

Key	Value type	Required	Comment
<same fields as in write request>			

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Key	Value type	Required	Comment
relay_mode	integer: 0, 1	y	Mode of external relay: • 0: Shorted • 1: Open
relay_status	integer: 0, 1	y	Status of external relay: • 0: Shorted • 1: Open

3.14 Heater / relay

Table 41 shows the endpoint references used to read and write heater relay settings.

Table 41: Heater relay settings – read / write

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/heater/relay	-	/heater/relay	/heater/relay	heater/relay/in heater/relay/out
write	/heater/relay/update	-	/heater/relay/update	/heater/relay/update	heater/relay/update/in heater/relay/update/out
subscription	/heater/relay/listen	-	-	/heater/relay/listen	heater/relay/listen

3.14.1 Read request

Table 42: Read request information

Key	Value type	Required	Comment
heater_nr	integer: 1...4	y	Heater number

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3.14.2 Write request

Table 43: Write request information

Key	Value type	Required	Comment
heater_nr	integer	y	Heater number (the same as in read request)
relay_mode	integer: 0, 1	y	Mode of external relay: • 0: Shorted • 1: Open

3.14.3 Response

Table 44: Response information

Key	Value type	Required	Comment
<same fields as in write request>			
relay_status	integer: 0, 1	y	Status of external relay: • 0: Shorted • 1: Open

3.15 Heater / output

Table 45 shows the endpoint references used to read heater relay settings.

Table 45: Heater relay settings – read / write

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/heater/output	-	-	-	heater/ output /in heater/ output /out
subscription	/heater/output/listen	-	-	-	heater/ output /listen

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3.16 Heater / historical data

Table 46 shows the endpoint reference for getting a heater's historical data from the device.

Table 46: Endpoint: Heater historical data

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/heater/historical-data	-	/heater/historical-data	/heater/historical-data	heater/historical-data/in heater/historical-data/out

3.16.1 Read request

Table 47: Read request information

Key	Value type	Required	Comment
heater_nr	integer: 1...4	y	Heater number
start_time	datetime string	y	Possible data formats are: • YYYY-MM-DD • YYYY-MM-DD HH:MM • YYYY-MM-DD HH:MM:SS • YYYY-MM-DDTHH:MM:SSZ
stop_time	datetime string	y	Same data formats as for start_time
fields	array of strings	y	Requested data fields (heater): • "power" • "current"

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3.16.2 Response

Table 48: Response information

Key	Value type	Required	Comment
<same fields as in write request>			
measurements	object with data arrays	y	Requested data Names of data array elements are the same as field names.
over_limit	boolean	y	Flag showing that the amount of requested data is too big.

Example

```
{
  "sender": "client-name-or-id",
  "hash": "this-is-just-hash-number-issued-by-client",
  "datetime": "2024-01-31T13:04:43.060313Z",
  "status": "OK",
  #
  "heater_nr": 1,
  "over_limit": False,
  "fields": [
    "timestamp",
    "power"
  ],
  "measurements": {
    "timestamp": [
      1542117983.719,
      ...
    ],
    "power": [
      1e-06,
      ...
    ]
  }
}
```


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3.17 Calibration curves (multiple)

Table 49 shows the endpoint references used to read information about calibration curves.

Table 49: Endpoint: Heater historical data

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/calibration-curves	/calibration-curves	-	/calibration-curves	calibration-curves/in calibration-curves/out
read (full data)	/calibration-curves/data	/calibration-curves/data	-	/calibration-curves/data	calibration-curves/data/in calibration-curves/data/out

3.17.1 Read request

Empty.

3.17.2 Response (no full calibration curves data)

There is an array of results for each calibration curve. Each element has the same format as shown in Table 50.

Table 50: Response information (no full calibration curves data)

Key	Value type	Required	Comment
calib_curve_nr	integer: 1...100	y	Calibration curve number
type	integer	y	Calibration curve type: • -1: Slot empty • 1: R → T curve • 2: X → T curve (8 Hz) • 3: X → T curve (64 Hz)
curve_hash	integer	y	This is an internal unique counter value for calibration curve

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3.17.3 Response (full calibration curves data)

There is an array of results for each calibration curve. Each element uses the same response format as shown in Table 55.

Example

```
  "sender": "client-name-or-id",
  "hash": "this-is-just-hash-number-issued-by-client",
  "datetime": "2024-01-31T13:04:43.060313Z",
  "status": "OK",
  #
  "data": [
    {
      "calib_curve_nr": 1,
      "type": -1,
      "curve_hash": -1
    },
    ...
    {
      "calib_curve_nr": 100,
      "type": 1,
      "curve_hash": -1
    }
  ]
}
```

3.18 Calibration curve (single)

Table 51 shows the endpoint references used to read and write single calibration curve data.

Table 51: Endpoint: Calibration curve (single)

Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
read	/calibration-curve	-	/calibration-curve	/calibration-curve	calibration-curve/in calibration-curve/out

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Requests	Endpoint	GET	POST	WEBSOCKET	MQTT
write	/calibration-curve/update	-	/calibration-curve/update	/calibration-curve/update	calibration-curve/update/in calibration-curve/update/out
subscription	/calibration-curve/listen	-	-	/calibration-curve/listen	/calibration-curve/listen
remove	/calibration-curve/remove	-	/calibration-curve/remove	/calibration-curve/remove	calibration-curve/remove/in calibration-curve/remove/out
file upload	/calibration-curve/file-upload	-	/calibration-curve/file-upload	/calibration-curve/file-upload	calibration-curve/file-upload/in calibration-curve/file-upload/out

3.18.1 Read request

Table 52: Read request information

Key	Value type	Required	Comment
calib_curve_nr	integer: 1...100	y	Calibration curve number

3.18.2 Write request

Table 53: Write request information

Key	Value type	Required	Comment
calib_curve_nr	integer: 1...100	y	Calibration curve number
name	string: max 50 chars	y	Name of the curve
sensor_model	string: max 50 chars	y	Sensor model
points	integer: 2...200	y	Number of points in the curve
impedances	array of floats: size max 200	y	Resistances or reactance in Ohm
temperatures	array of floats: size max 200	y	Temperatures in kelvin. Same array size as with impedances.

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Key	Value type	Required	Comment
type	integer	y	Calibration curve type: • 1: Slot empty (only used in response) • 1: R → T curve • 2: X → T curve (8 Hz) • 3: X → T curve (64 Hz)

3.18.3 Remove request

A remove request has the same format shown in “Read request”.

3.18.4 Upload request

Table 54: Upload request information

Key	Value type	Required	Comment
calib_curve_nr	integer: 1...100	y	Calibration curve number
file_contents	string	y	Contents of calibration curve file in string format Used only in write request message

3.18.5 Response

Table 55: Response information

Key	Value type	Required	Comment
<same fields as in write request>			
curve_hash	integer	y	This is an internal unique counter value for calibration curve

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Example

```
{
  "sender": "client-name-or-id",
  "hash": "this-is-just-hash-number-issued-by-client",
  "datetime": "2024-01-31T13:04:43.060313Z",
  "status": "OK",
  #
  "calib_curve_nr": 4,
  "name": "NAME",
  "sensor_model": "MODEL",
  "points": 4,
  "impedances": [0.001, 0.002, 0.003, 0.004],
  "temperatures": [1e-06, 2e-06, 3e-06, 4e-06],
  "type": 1,
  "curve_hash": 1
}
```


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4 Examples

This chapter shows a few examples where the API is written in Python language. The examples assume that the following Python packages are installed on the system.

```
requests>=0.19.1
websocket-client>=0.53.0
```

4.1 Read values with HTTP GET

This program is used to read values continuously using the /channel/measurement/latest endpoint.

```
# -*- coding: utf-8 -*-
"""
Example: reading current measurement results using HTTP interface.
"""

# --- Imports

#
import json
import requests
from time import sleep

# --- Setup

# Define your device IP here
DEVICE_IP = 'localhost'

# Timeout for get/post operations (in seconds)
TIMEOUT = 10

# Timeout used for loop cycle (in seconds)
LOOP_TIMEOUT = 10

# --- Read measurement data in the loop

#
url = 'http://{}:5001/channel/measurement/latest'.format(DEVICE_IP)

#
while True:
    req = requests.get(url, timeout=TIMEOUT)
    data = req.json()

    #
    print('Response: \n{}'.format(json.dumps(data, indent=2)))

    #
    sleep(LOOP_TIMEOUT)
```


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4.2 Read values with WebSocket

This Python code sample reads the latest measurement results with a WebSocket interface and the /channel/measurement/listen endpoint.

```
# -*- coding: utf-8 -*-
"""
Example: reading current measurement results using WebSocket interface.
"""

# --- Imports

#
import json
import websocket

# --- Setup

# Define your device IP here
DEVICE_IP = 'localhost'

# Timeout for WebSocket operations (in seconds)
TIMEOUT = 10

# --- Listen for measurement data updates

#
url = 'ws://{}:5002/channel/measurement/listen'.format(DEVICE_IP)
ws = websocket.create_connection(url, timeout=TIMEOUT)
#
while True:
    try:
        resp = ws.recv()
    except websocket.WebSocketTimeoutException:
        continue

    data = json.loads(resp)

    #
    print('Response: \n{}'.format(json.dumps(data, indent=2)))
```

4.3 Read settings

This example reads settings using the endpoints /system and /channel.

```
# -*- coding: utf-8 -*-
"""
Example: read different settings using GET and POST methods of HTTP
protocol.
"""

# --- Imports

import json
import requests

# --- Setup

# Define your device IP here
DEVICE_IP = 'localhost'

# Timeout for http operations (in seconds)
TIMEOUT = 10

# --- Part 1: Read system settings using HTTP GET request
```


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```
#
url = 'http://{}:5001/system'.format(DEVICE_IP)

#
req = requests.get(url, timeout=TIMEOUT)
data = req.json()

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))

# --- Part 2: Read channel N settings using HTTP POST request

#
url = 'http://{}:5001/channel'.format(DEVICE_IP)

data = {
    'channel_nr': 2
}
#
req = requests.post(url, json=data, timeout=TIMEOUT)
data = req.json()

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))
```

4.4 Write settings

This example writes settings using the endpoint /channel/update.

```
# -*- coding: utf-8 -*-
"""
Example: update some of channel settings using HTTP interface.
"""

# --- Imports

import json
import requests

# --- Setup

# Define your device IP here
DEVICE_IP = 'localhost'

# Timeout for http operations (in seconds)
TIMEOUT = 10

# --- Update settings

#
url = 'http://{}:5001/channel/update'.format(DEVICE_IP)

data = {
    'channel_nr': 1,
    'excitation_mode': 0,
    'excitation_current_range': 11,
    'calib_curve_nr': 22
}

#
req = requests.post(url, json=data, timeout=TIMEOUT)
data = req.json()

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))
```


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4.5 Read historical data

This example loads old measurement results from the device using the endpoint /channel/historical-data.

```
# -*- coding: utf-8 -*-
"""
Example: read channel old measurement data.
"""

# --- Imports

#
import json
import requests

# --- Setup

# Define your device IP here
DEVICE_IP = 'localhost'

# Timeout for get/post operations (in seconds)
TIMEOUT = 100

# --- Read channel N old measurement data in defined time period

#
url = 'http://{}:5001/channel/historical-data'.format(DEVICE_IP)

data = {
    'channel_nr': 8,
    'start_time': '2023-01-31T00:00:00Z',
    'stop_time': '2024-01-31T00:00:00Z',
    'fields': [
        'temperature',
        'resistance',
        'reactance'
    ]
}

#
req = requests.post(url, json=data, timeout=TIMEOUT)
data = req.json()

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))
```

4.6 Calibration curve loading

This example stores a calibration curve to the device using the endpoint /calibration-curve.

```
# -*- coding: utf-8 -*-
"""
Example: calibration curve manipulation.
"""

# --- Imports

#
import json
import requests

# --- Setup
```


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```
# Define your device IP here
DEVICE_IP = 'localhost'

# Timeout for http operations (in seconds)
TIMEOUT = 10

# Calibration curve number used in example
CALIB_CURVE_NR = 98

# --- Read calibration curve N data

#
url = 'http://{}:5001/calibration-curve'.format(DEVICE_IP)

#
data = {
    'calib_curve_nr': CALIB_CURVE_NR
}

#
req = requests.post(url, json=data, timeout=TIMEOUT)
data = req.json()

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))

# --- Update calibration curve N data

url = 'http://{}:5001/calibration-curve/update'.format(DEVICE_IP)

#
data = {
    'calib_curve_nr': CALIB_CURVE_NR,
    'name': 'NAME',
    'sensor_model': 'MODEL',
    'points': 2,
    'impedances': [1, 2],
    'temperatures': [1, 2],
    'type': 1
}

#
req = requests.post(url, json=data, timeout=TIMEOUT)
data = req.json()

#
print('Response: \n{}'.format(json.dumps(data, indent=2)))
```


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